Multiple-choice questions

|!|BCEMC0230200001|!|

Which of the following is/are the sign(s) of influenza?

(1) fever
(2) sore throat
(3) fatigue

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0230400002|!|

Which of the following disease(s) can be transmitted to other people when an infected person coughs without covering his mouth and nose?

(1) AIDS
(2) tuberculosis
(3) influenza

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0230400003|!| ★

Which of the following statements about a prevention measure against dengue fever is correct?

A. Wear a condom during sexual contact as the pathogens that cause dengue fever can be transmitted by body fluids.
B. Wear long-sleeved clothing when staying outdoors as mosquito bites can transmit the pathogens that cause dengue fever.
C. Cook food thoroughly as the pathogens that cause dengue fever can multiply in undercooked food.
D. Avoid going to crowded public places as the pathogens that cause dengue fever can be transmitted through air or droplets.

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230400004|!|

Which of the following can reduce the risk of humans contracting avian flu?

(1) take antibiotics
(2) avoid touching live chickens
(3) avoid eating raw eggs

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0230300005|!|

Which of the following infectious diseases is caused by a virus?

A. cholera
B. tuberculosis
C. athlete's foot
D. measles

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230400006|!|

Which of the following diseases ***cannot*** be transmitted by mosquitoes?

A. dengue fever
B. Japanese encephalitis
C. cholera
D. malaria

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0230500007|!| ★

✂

Which of the following diseases can be treated by antibiotics?

A. hepatitis A
B. tuberculosis
C. AIDS
D. dengue fever

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0230200008|!|

Which of the following is an infectious disease?

A. oesteoporosis
B. hand, foot and mouth disease
C. lung cancer
D. diabetes mellitus

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0230200009|!| ★

How may HIV be passed from an infected mother to her baby?

(1) through contact with the mother's blood and vaginal fluids during childbirth
(2) through hugging
(3) through breast milk

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0230300010|!|

Which of the following descriptions about the pathogens that cause malaria is/are correct?

(1) They are mosquitoes.
(2) They are eukaryotes.
(3) They invade liver cells and red blood cells.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230300011|!|

Which of the following combinations correctly shows the types of pathogens that cause avian flu and malaria?

	Avian flu	***Malaria***
A.	bacteria	protists
B.	bacteria	bacteria
C.	viruses	protists
D.	viruses	bacteria

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0230400012|!| ★

Which of the following diseases' transmission link can be broken by eliminating mosquitoes?

(1) dengue fever
(2) AIDS
(3) Japanese encephalitis

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0230400013|!|

Which of the following combinations ***incorrectly*** matches the disease and its way of transmission?

	Disease	***Way of transmission***
A.	hepatitis B	by contaminated food
B.	tuberculosis	by air and droplets
C.	syphilis	by body fluids
D.	cholera	by contaminated water

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230400014|!|

Which of the following can help control the spread of influenza?

(1) maintain good ventilation indoors
(2) wear a surgical mask if infected
(3) avoid close contact with infected people

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0230400015|!| ★

In 2003, there was an outbreak of SARS (severe acute respiratory syndrome) in Hong Kong. Which of the following led to the quick spread of SARS at that time?

(1) frequent migrations of people around the world
(2) worsening air pollution problems
(3) transmission of SARS virus through droplets

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0230400016|!| ★★ SI

In the 19th century, many women died from infections after giving birth in hospitals. A doctor observed the death rate in two maternity wards in the same hospital for a year and collected the data shown in the table.

	Ward A	Ward B
Number of women who gave birth	202	198
Number of women who died from infections after giving birth	67	19

The doctor also noticed that medical students were delivering babies in ward A after they had been dissecting corpses as part of their studies, while only midwives delivered babies in ward B.

Based on the above information, what is the ***best*** explanation for the higher death rate in ward A?

A. Ward A was more crowded than ward B.
B. Midwives were more skilful in delivering babies than medical students.
C. The higher death rate in ward A occurred by chance.
D. Medical students carried pathogens from corpses to the women in labour.

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0230400017|!|

Which of the following measures can protect oneself from athlete's foot?

A. apply deodorant to the feet
B. avoid visiting crowded places
C. avoid sharing needles or syringes with others
D. avoid sharing shoes and socks with others

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230300018|!|

Severe acute respiratory syndrome (SARS) and avian flu are diseases caused by

A. bacteria.
B. fungi.
C. viruses.
D. protists.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230400019|!|

Which of the following diseases is/are transmitted by body fluids?

(1) hepatitis A
(2) hepatitis B
(3) AIDS

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

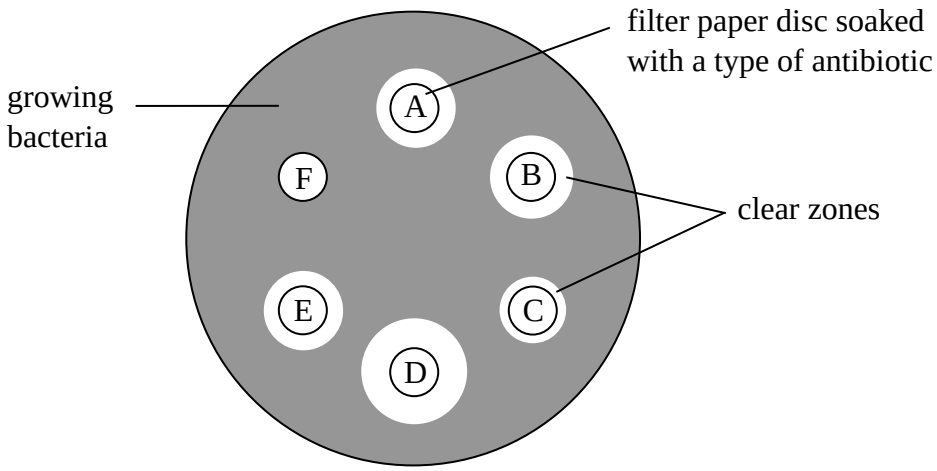
Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0230500020|!| ★ SI

✂

In an investigation, scientists studied the effect of different antibiotics on a bacterial species. The following diagram shows their results.



Which of the following can be concluded from the results?

A. This bacterial species is resistant to antibiotic F.
B. Antibiotic F is the most effective antibiotic against this bacterial species.
C. Antibiotic B is less effective against this bacterial species than antibiotic C.
D. This bacterial species is resistant to all of the six antibiotics.

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0230500021|!|

✂

Which of the following speed up the emergence of antibiotic-resistant bacteria?

(1) Patients do not finish the whole course of antibiotic prescription.
(2) Most patients are immuno-deficient.
(3) Doctors often prescribe antibiotics for most patients.

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230500022|!|

✂

Which of the following is ***not*** a mechanism of antibiotics in inhibiting the growth of bacteria?

A. inhibiting the replication of bacterial DNA
B. destroying bacterial cell membranes
C. inhibiting the synthesis of bacterial cell walls
D. facilitating the phagocytosis of bacteria

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0230500023|!|

✂

Which of the following correctly describes the action of sulpha drugs against bacteria?

A. Sulpha drugs inhibit the synthesis of folic acid in bacteria.
B. Sulpha drugs inhibit the synthesis of amino acid in bacteria.
C. Sulpha drugs inactivate all enzymes in bacteria.
D. Sulpha drugs inhibit the replication of bacterial DNA.

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0230500024|!|

✂

Which of the following statements about penicillin is/are correct?

(1) It is produced by a fungus called Penicillium.
(2) It can be used to treat all types of infections.
(3) It kills pathogenic bacteria but not beneficial bacteria.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0230500025|!| ★

✂

After the application of antibiotics to treat a disease, a higher percentage of disease-causing bacteria show resistance to the antibiotics. Which of the following statements correctly explains this phenomenon?

A. Non-resistant bacteria become resistant after being exposed to the antibiotics.
B. Resistant bacteria survive during the treatment and pass on the resistance character to the offspring.
C. Non-resistant bacteria alter the metabolic pathways in the patients' cells and develop resistance.
D. Beneficial bacteria are killed by antibiotics and this encourages the growth of resistant bacteria.

□

##B##

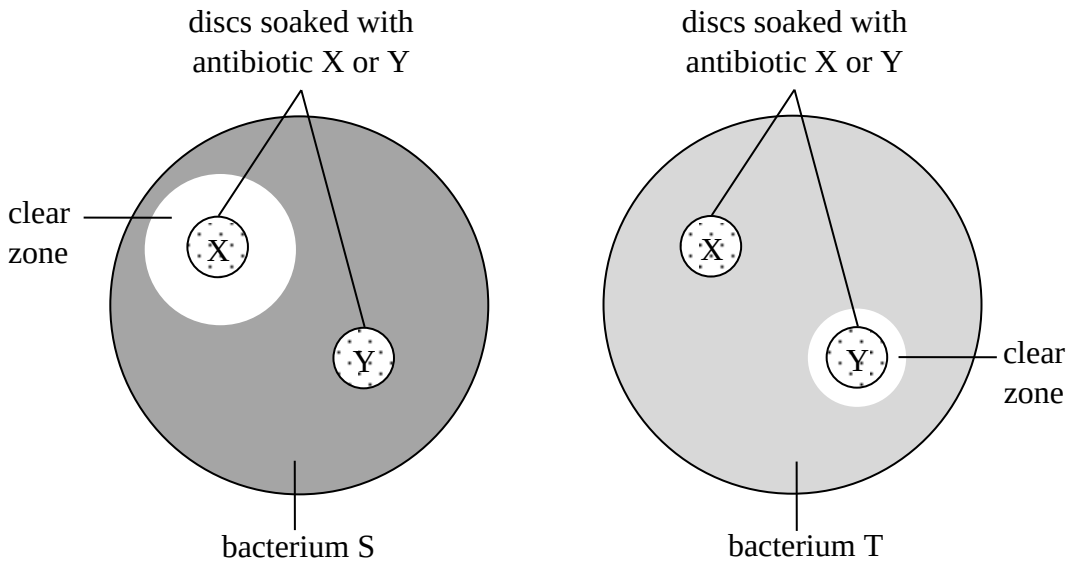
Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0230500026|!| ★ **SI**

✂

The diagram below shows two agar plates, each with two different antibiotic discs (X and Y). Each agar plate has a different species of bacterium (S or T) growing on it.



Which of the following conclusions can be drawn based on the results shown?

(1) The growth of bacterium S was inhibited by antibiotic X but not antibiotic Y.
(2) Antibiotic X was effective for inhibiting the growth of bacteria S and T.
(3) Bacterium T was resistant to antibiotic X.

A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

□

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0230500027|!| ★

✂

Why are antibiotics effective for killing bacterial cells but not viruses?

A. Viruses are not pathogens.
B. Viruses can destroy antibiotics.
C. Viruses do not have genetic materials.
D. Viruses do not have metabolism like bacterial cells do.

☐

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0230500028|!| ★

✂

Which of the following statements about antibiotics and sulpha drugs is correct?

A. Both antibiotics and sulpha drugs are naturally produced by microorganisms.
B. Antibiotics and sulpha drugs can be used to treat bacterial infections.
C. Antibiotics and sulpha drugs work by inhibiting the synthesis of nucleic acids.
D. Bacteria may develop resistance against antibiotics but not sulpha drugs.

☐

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0230500029|!|

✂

Which of the following can be used to treat bacterial infections?

A. insulin injections
B. disinfectants
C. vaccines
D. sulpha drugs

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0230500030|!| ★

✂

Which of the following is/are ***not*** the proper way(s) of using antibiotics?

(1) Finishing the whole course of antibiotic prescription even after symptoms have subsided
(2) Adding antibiotics to animal feed to prevent illnesses among livestock
(3) Prescribing several types of antibiotics to patients whenever a bacterial infection is diagnosed

A. (1) only
B. (2) only
C. (3) only
D. (2) and (3) only

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

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